

AUTONOMOUS DRONE NAVIGATION

GPS-Denied Coordinate System Approach

1. PROJECT IDEA

The core objective of this project is to develop a fully autonomous drone navigation system capable of operating in entirely GPS-denied environments. Instead of relying on satellite-based global positioning, the drone will navigate using an internal, self-generated relative coordinate system (X, Y, Z axes mapped from its initial takeoff point).

By leveraging an onboard flight controller integrated with a fusion of local sensors—specifically an Inertial Measurement Unit (IMU) for acceleration and rotation, and an Optical Flow sensor or lightweight monocular camera for velocity estimation—the drone will continuously calculate its current coordinates through advanced dead reckoning and state estimation algorithms. The system will accept relative target coordinates as inputs, autonomously compute the optimal flight path, and navigate to the destination while adjusting for drift and environmental obstacles, completely independent of external GPS signals.

2. BACKGROUND

Unmanned Aerial Vehicles (UAVs) have traditionally depended on the Global Navigation Satellite System (GNSS/GPS) to determine their absolute position, altitude, and velocity. GPS provides a reliable global framework that makes waypoint navigation relatively straightforward. However, GPS technology has fundamental physical and security limitations. The signals transmitted from satellites are inherently weak by the time they reach Earth's surface.

Consequently, GPS signals are easily blocked by physical barriers, a phenomenon known as signal attenuation or multi-pathing. This makes GPS useless in environments such as dense urban areas with tall skyscrapers (urban canyons), deep forest canopies, underground mines, and indoor facilities. Furthermore, GPS signals are highly susceptible to malicious interference, including RF jamming and spoofing, which can disorient drones and cause catastrophic mission failures. As drones are increasingly deployed in these complex, obscured, or contested environments, the reliance on a single, external positioning source has become a critical bottleneck in UAV evolution.

3. SIGNIFICANCE

Developing a robust, GPS-independent navigation system is not just an academic exercise; it is a vital requirement for the next generation of autonomous robotics. The significance of this project lies in its potential to unlock new, critical use cases for drones.

Why is it Required?

A relative coordinate navigation system is required because absolute reliance on external infrastructure (like satellites) creates a single point of failure. In emergency scenarios or secure operations, operators cannot wait for satellite acquisition or risk signal loss mid-flight. The drone must possess the internal "intelligence" and sensor fusion capabilities to understand its spatial movement independently to guarantee mission completion and vehicle recovery.

How is it Useful? (Real-world Applications)

- **Search and Rescue (SAR) in Disaster Zones:** Drones can autonomously navigate inside collapsed buildings, caves, or earthquake rubble where GPS cannot penetrate, mapping the area and locating survivors using relative coordinate waypoints.
- **Underground & Industrial Operations:** Enables automated inspections in GPS-devoid areas like underground mining shafts, sewer networks, and the interiors of large manufacturing plants or warehouses for inventory management.
- **Defense and Tactical Operations:** In electronic warfare (EW) environments where adversaries actively jam or spoof GPS signals, a coordinate-based internal navigation system ensures that tactical UAVs remain operational and can execute their flight paths safely.

4. LITERATURE SURVEY

To design an effective GPS-denied navigation framework, it is crucial to analyze existing methodologies in academic literature. The following survey explores key papers and prevailing techniques—ranging from Visual SLAM to Visual-Inertial Odometry—evaluating their strengths and identifying gaps that our project aims to address.

Review 1: Visual-Inertial Odometry (VIO) for State Estimation

Based on methodologies from: "VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator" (Qin et al.)

Methodology: This approach tightly couples pre-integrated IMU measurements with feature observations from a single camera. It creates a robust state estimator that calculates the drone's position, velocity, and orientation in a local coordinate frame.

Relevance to Project: VIO is highly relevant because it is computationally efficient and requires minimal hardware (just a camera and an IMU). This supports our goal of a lightweight system. However, pure VIO can accumulate drift over long distances, indicating that our system must implement a robust error-correction loop or limit flights to structured coordinate matrices.

Review 2: Vision-Based SLAM Systems

Based on methodologies from: "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM" (Campos et al.)

Methodology: ORB-SLAM3 extracts features from camera feeds to build a sparse 3D map of the environment while simultaneously determining the drone's location within that map. It excels at "loop closure"—recognizing a previously visited area to correct accumulated navigational drift.

Relevance to Project: While SLAM offers exceptional accuracy for local coordinate mapping, it requires significant onboard computational power (e.g., companion computers like Jetson Nano), which drains battery and adds weight. Our project will aim for a more streamlined approach, prioritizing trajectory execution over heavy 3D map generation.

Review 3: LiDAR Odometry in Denied Environments

Based on methodologies from: "LOAM: Lidar Odometry and Mapping in Real-time" (Zhang & Singh)

Methodology: LOAM utilizes 2D or 3D LiDAR sensors to bounce laser pulses off surroundings, calculating time-of-flight to generate highly accurate point clouds. By matching sequential point clouds, the drone deduces its motion vector and coordinates.

Relevance to Project: LiDAR is completely independent of lighting conditions, unlike camera-based systems. However, traditional LiDAR sensors are heavy and expensive. This literature suggests that if we incorporate spatial mapping, a 1D or 2D lightweight LiDAR (for altitude and obstacle avoidance) combined with optical flow is more feasible for a standard UAV payload than full 3D LOAM.

Review 4: Optical Flow for Velocity Estimation

Based on methodologies from: "Optical Flow based Autonomous Hovering and Navigation" (Various IEEE frameworks)

Methodology: Optical flow sensors (often downward-facing) calculate the rate at which pixels move across the sensor's field of view. When combined with a rangefinder (to determine altitude), it provides accurate planar (X, Y) velocity without needing to recognize specific environmental features.

Relevance to Project: This forms the bedrock of our proposed solution. By relying on optical flow for short-term X/Y movement and an IMU for attitude, we can create a highly efficient relative coordinate system. The literature confirms this method is extremely lightweight, though it requires a textured ground surface to function optimally.

Synthesis & Conclusion of Survey

The literature reveals a distinct trade-off in GPS-denied navigation: high accuracy typically demands heavy computational overhead (SLAM) or heavy physical payloads (LiDAR). Furthermore, systems like UWB require external infrastructure, which defeats the purpose of autonomous exploration in unknown environments. Our project addresses these gaps by adopting a hybrid approach: utilizing the lightweight principles of Optical Flow and IMU dead-reckoning (inspired by VIO) to track a self-contained relative coordinate system. This ensures the drone remains agile, computationally efficient, and completely independent of external signals or pre-installed infrastructure.